Communications Relay: System Design Description

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Revision	C
Status	APPROVED
Classification	INTERNAL
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1. System overview

Communications covers HoloNet transceivers, comm array dishes and the switches that route passenger, cargo and station traffic.

The system is led by Computing & Communications. It is delivered through 5 work packages with a combined budget of 70,982.43 Mcr.

2. Components

Code	Component	Specification	Drawing	Bill of materials
CM-HNT	HoloNet transceiver	SHV-CM-SPEC-0101	SHV-CM-DWG-0201	SHV-CM-BOM-0301
CM-CMA	Comm array dish	SHV-CM-SPEC-0102	SHV-CM-DWG-0202	SHV-CM-BOM-0302
CM-TRS	Traffic routing switch	SHV-CM-SPEC-0103	SHV-CM-DWG-0203	SHV-CM-BOM-0303

3. Architecture

3.1 HoloNet transceiver

The holonet transceiver is specified in SHV-CM-SPEC-0101. Its governing parameter is the channels, set at 1,146 ea. The responsible engineer is Kaspar Marlow.

3.2 Comm array dish

The comm array dish is specified in SHV-CM-SPEC-0102. Its governing parameter is the dish diameter, set at 112 m. The responsible engineer is Vira Renner.

3.3 Traffic routing switch

The traffic routing switch is specified in SHV-CM-SPEC-0103. Its governing parameter is the throughput, set at 8.2 Tbit/s. The responsible engineer is Edda Palin.

4. Interfaces

Document	Title
SHV-ICD-CN-CM	Interface Control Document: Computer Core & Data Network / Communications Relay

5. Work packages

Work package	Name	Owner	Status	Complete
WP-SHV-CM-01	Communications Relay: Design & analysis	Pax Zander	IN_PROGRESS	84.8%

Work package	Name	Owner	Status	Complete
WP-SHV-CM-02	Communications Relay: Fabricate & install holonet transceiver	Pax Zander	IN_PROGRESS	89.7%
WP-SHV-CM-03	Communications Relay: Fabricate & install comm array dish	Vance Coldbrook	IN_PROGRESS	75.2%
WP-SHV-CM-04	Communications Relay: Fabricate & install traffic routing switch	Rafe Craddock	IN_PROGRESS	67.7%
WP-SHV-CM-05	Communications Relay: Integration & test	Vira Renner	IN_PROGRESS	85.1%

6. Risks

- Supply of holonet transceiver assemblies depends on a single supplier.
- Supply of comm array dish assemblies depends on a single supplier.
- Integration slots in the station schedule are shared with neighbouring zones.